

Java ArrayList

20 Practice Questions — Pseudocode, Code, Dry Run & Output

Each question includes the problem statement, pseudocode, complete working Java code, a step-by-step dry run table, and the final output — companion to the ArrayList class notes.

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Question 1: Create an ArrayList and Print It

Problem

Create an ArrayList of integers [5, 10, 15, 20] and print it.

Pseudocode

```
FUNCTION createAndPrint():  
  list = new ArrayList  
  ADD 5, 10, 15, 20 to list  
  PRINT list
```

Java Code

```
static void createAndPrint() {  
  List<Integer> list = new ArrayList<>();  
  list.add(5); list.add(10); list.add(15); list.add(20);  
  System.out.println(list);  
}  
public static void main(String[] args) {  
  createAndPrint();  
}
```

Dry Run

Call	list after
add(5)	[5]
add(10)	[5, 10]
add(15)	[5, 10, 15]
add(20)	[5, 10, 15, 20]

Output: [5, 10, 15, 20]

Question 2: Find the Sum and Average of an ArrayList

Problem

Given [4, 8, 15, 16, 23, 42], compute the sum and average using an ArrayList.

Pseudocode

```
FUNCTION sumAvg(list):  
  sum = 0  
  FOR each v in list: sum = sum + v  
  avg = sum / list.size()  
  RETURN (sum, avg)
```

Java Code

```
static void sumAvg(List<Integer> list) {  
  int sum = 0;  
  for (int v : list) sum += v;  
  double avg = (double) sum / list.size();  
  System.out.println("Sum=" + sum + " Avg=" + avg);  
}  
public static void main(String[] args) {  
  List<Integer> list = new ArrayList<>(List.of(4, 8, 15, 16, 23, 42));  
  sumAvg(list);  
}
```

Dry Run

v	running sum
4	4

v	running sum
8	12
15	27
16	43
23	66
42	108

Output: Sum=108 Avg=18.0

Question 3: Find Maximum and Minimum in an ArrayList

Problem

Find max and min in the ArrayList [7, 2, 9, 4, 11, 3].

Pseudocode

```
FUNCTION findMinMax(list):
  max = list.get(0), min = list.get(0)
  FOR i FROM 1 TO list.size() - 1:
    IF list.get(i) > max: max = list.get(i)
    IF list.get(i) < min: min = list.get(i)
  RETURN (min, max)
```

Java Code

```
static void findMinMax(List<Integer> list) {
  int max = list.get(0), min = list.get(0);
  for (int i = 1; i < list.size(); i++) {
    int v = list.get(i);
    if (v > max) max = v;
    if (v < min) min = v;
  }
  System.out.println("Min=" + min + " Max=" + max);
}

public static void main(String[] args) {
  List<Integer> list = new ArrayList<>(List.of(7, 2, 9, 4, 11, 3));
  findMinMax(list);
}
```

Dry Run

i	list.get(i)	max	min
-	-	7	7
1	2	7	2
2	9	9	2
3	4	9	2
4	11	11	2
5	3	11	2

Output: Min=2 Max=11

Question 4: Remove Duplicate Elements from an ArrayList

Problem

Remove duplicates from [1, 3, 2, 3, 4, 1, 5] while keeping the first occurrence order.

Pseudocode

```
FUNCTION removeDuplicates(list):  
  seen = empty LinkedHashSet  
  FOR each v in list: ADD v to seen  
  RETURN new ArrayList from seen
```

Java Code

```
static List<Integer> removeDuplicates(List<Integer> list) {  
  Set<Integer> seen = new LinkedHashSet<>(list);  
  return new ArrayList<>(seen);  
}  
public static void main(String[] args) {  
  List<Integer> list = new ArrayList<>(List.of(1, 3, 2, 3, 4, 1, 5));  
  System.out.println(removeDuplicates(list));  
}
```

Dry Run

v	seen.add(v)	seen after
1	true	{1}
3	true	{1, 3}
2	true	{1, 3, 2}
3	false (dup)	{1, 3, 2}
4	true	{1, 3, 2, 4}
1	false (dup)	{1, 3, 2, 4}
5	true	{1, 3, 2, 4, 5}

Output: [1, 3, 2, 4, 5]

Question 5: Reverse an ArrayList

Problem

Reverse the ArrayList [10, 20, 30, 40, 50] in place.

Pseudocode

```
FUNCTION reverseList(list):  
  CALL Collections.reverse(list) // or two-pointer swap manually
```

Java Code

```
static void reverseList(List<Integer> list) {  
  Collections.reverse(list);  
}  
public static void main(String[] args) {  
  List<Integer> list = new ArrayList<>(List.of(10, 20, 30, 40, 50));  
  reverseList(list);  
  System.out.println(list);  
}
```

Dry Run

left	right	list state
0	4	[50, 20, 30, 40, 10]

left	right	list state
1	3	[50, 40, 30, 20, 10]
2	2	stop (left==right)

Output: [50, 40, 30, 20, 10]

Question 6: Remove All Even Numbers from an ArrayList

Problem

From [1, 2, 3, 4, 5, 6, 7, 8], remove all even numbers.

Pseudocode

```
FUNCTION removeEvens(list):
  list.removeIf(v -> v MOD 2 == 0)
```

Java Code

```
static void removeEvens(List<Integer> list) {
  list.removeIf(v -> v % 2 == 0);
}
public static void main(String[] args) {
  List<Integer> list = new ArrayList<>(List.of(1, 2, 3, 4, 5, 6, 7, 8));
  removeEvens(list);
  System.out.println(list);
}
```

Dry Run

v	v % 2 == 0?	kept?
1	no	yes
2	yes	removed
3	no	yes
4	yes	removed
5	no	yes
6	yes	removed
7	no	yes
8	yes	removed

Output: [1, 3, 5, 7]

Question 7: Find the Second Largest Element in an ArrayList

Problem

Find the second largest value in [12, 35, 1, 10, 34, 1].

Pseudocode

```
FUNCTION secondLargest(list):
  largest = -infinity, second = -infinity
  FOR each v in list:
    IF v > largest: second = largest, largest = v
    ELSE IF v > second AND v != largest: second = v
  RETURN second
```

Java Code

```
static int secondLargest(List<Integer> list) {
  int largest = Integer.MIN_VALUE, second = Integer.MIN_VALUE;
  for (int v : list) {
```

```

if (v > largest) {
    second = largest;
    largest = v;
} else if (v > second && v != largest) {
    second = v;
}
}
return second;
}
public static void main(String[] args) {
    List<Integer> list = new ArrayList<>(List.of(12, 35, 1, 10, 34, 1));
    System.out.println(secondLargest(list));
}

```

Dry Run

v	largest	second
12	12	MIN
35	35	12
1	35	12
10	35	12
34	35	34
1	35	34

Output: 34

Question 8: Merge Two ArrayLists into One (No Duplicates)

Problem

Merge [1, 2, 3] and [2, 3, 4] into a single ArrayList with unique values.

Pseudocode

```
FUNCTION mergeUnique(list1, list2):  
    result = new LinkedHashSet  
    ADD all of list1 to result  
    ADD all of list2 to result  
    RETURN new ArrayList from result
```

Java Code

```
static List<Integer> mergeUnique(List<Integer> list1, List<Integer> list2) {  
    Set<Integer> result = new LinkedHashSet<>(list1);  
    result.addAll(list2);  
    return new ArrayList<>(result);  
}  
  
public static void main(String[] args) {  
    List<Integer> a = List.of(1, 2, 3), b = List.of(2, 3, 4);  
    System.out.println(mergeUnique(a, b));  
}
```

Dry Run

Step	result after
add all of list1	{1,2,3}
add all of list2 (2,3 already present)	{1,2,3,4}

Output: [1, 2, 3, 4]

Question 9: Find the Intersection of Two ArrayLists

Problem

Find common elements between [1, 2, 2, 3, 4] and [2, 2, 4, 6, 7].

Pseudocode

```
FUNCTION intersection(list1, list2):  
    set1 = distinct elements of list1  
    result = empty LinkedHashSet  
    FOR each v in list2:  
        IF v IN set1: ADD v to result  
    RETURN new ArrayList from result
```

Java Code

```
static List<Integer> intersection(List<Integer> list1, List<Integer> list2) {  
    Set<Integer> set1 = new HashSet<>(list1);  
    Set<Integer> result = new LinkedHashSet<>();  
    for (int v : list2) {  
        if (set1.contains(v)) result.add(v);  
    }  
    return new ArrayList<>(result);  
}  
  
public static void main(String[] args) {  
    List<Integer> a = List.of(1, 2, 2, 3, 4), b = List.of(2, 2, 4, 6, 7);  
    System.out.println(intersection(a, b));  
}
```

Dry Run

v (from list2)	in set1?	result after
2	yes	{2}
2	yes (dup ignored)	{2}
4	yes	{2, 4}
6	no	{2, 4}
7	no	{2, 4}

Output: [2, 4]

Question 10: Sort an ArrayList of Strings by Length

Problem

Sort ["banana", "kiwi", "apple", "fig"] by string length, ascending.

Pseudocode

```
FUNCTION sortByLength(list):
  list.sort(Comparator.comparingInt(String::length))
```

Java Code

```
static void sortByLength(List<String> list) {
  list.sort(Comparator.comparingInt(String::length));
}
public static void main(String[] args) {
  List<String> list = new ArrayList<>(List.of("banana", "kiwi", "apple", "fig"));
  sortByLength(list);
  System.out.println(list);
}
```

Dry Run

word	length
fig	3
kiwi	4
apple	5
banana	6

Output: [fig, kiwi, apple, banana]

Question 11: Rotate an ArrayList to the Right by K Positions

Problem

Rotate [1, 2, 3, 4, 5] to the right by k = 2 using Collections.rotate.

Pseudocode

```
FUNCTION rotateRight(list, k):
  CALL Collections.rotate(list, k)
```

Java Code

```
static void rotateRight(List<Integer> list, int k) {
  Collections.rotate(list, k);
}
public static void main(String[] args) {
  List<Integer> list = new ArrayList<>(List.of(1, 2, 3, 4, 5));
  rotateRight(list, 2);
  System.out.println(list);
}
```

Dry Run

Step	list state
initial	[1, 2, 3, 4, 5]
rotate by 2 (last 2 move to front)	[4, 5, 1, 2, 3]

Output: [4, 5, 1, 2, 3]

Question 12: Find All Pairs in an ArrayList that Sum to a Target

Problem

Given [8, 7, 2, 5, 3, 1] and target 10, find all pairs summing to 10.

Pseudocode

```
FUNCTION findPairs(list, target):  
  seen = empty HashSet  
  FOR each v in list:  
    complement = target - v  
    IF complement IN seen: PRINT (complement, v)  
    ADD v to seen
```

Java Code

```
static void findPairs(List<Integer> list, int target) {  
  Set<Integer> seen = new HashSet<>();  
  for (int v : list) {  
    int complement = target - v;  
    if (seen.contains(complement)) {  
      System.out.println("(" + complement + ", " + v + ")");  
    }  
    seen.add(v);  
  }  
}  
  
public static void main(String[] args) {  
  List<Integer> list = List.of(8, 7, 2, 5, 3, 1);  
  findPairs(list, 10);  
}
```

Dry Run

v	complement	in seen?
8	2	no
7	3	no
2	8	yes -> (8, 2)
5	5	no
3	7	yes -> (7, 3)
1	9	no

Output: (8, 2)\n(7, 3)

Question 13: Convert ArrayList of Integers to Array

Problem

Convert [3, 6, 9, 12] from an ArrayList to an int[] array.

Pseudocode

```
FUNCTION toArray(list):  
  arr = new int array of size list.size()  
  FOR i FROM 0 TO list.size() - 1:  
    arr[i] = list.get(i)  
  RETURN arr
```

Java Code

```
static int[] toIntArray(List<Integer> list) {  
  int[] arr = new int[list.size()];  
  for (int i = 0; i < list.size(); i++) {  
    arr[i] = list.get(i);  
  }  
  return arr;  
}  
  
public static void main(String[] args) {  
  List<Integer> list = List.of(3, 6, 9, 12);  
  System.out.println(Arrays.toString(toIntArray(list)));  
}
```

Dry Run

i	list.get(i)	arr after
0	3	[3, 0, 0, 0]
1	6	[3, 6, 0, 0]
2	9	[3, 6, 9, 0]
3	12	[3, 6, 9, 12]

Output: [3, 6, 9, 12]

Question 14: Check if an ArrayList is a Palindrome

Problem

Check whether [1, 2, 3, 2, 1] reads the same forwards and backwards.

Pseudocode

```
FUNCTION isPalindrome(list):  
  left = 0, right = list.size() - 1  
  WHILE left < right:  
    IF list.get(left) != list.get(right): RETURN false  
    left++, right--  
  RETURN true
```

Java Code

```
static boolean isPalindrome(List<Integer> list) {  
  int left = 0, right = list.size() - 1;  
  while (left < right) {  
    if (!list.get(left).equals(list.get(right))) return false;  
    left++; right--;  
  }  
  return true;  
}  
  
public static void main(String[] args) {  
  List<Integer> list = List.of(1, 2, 3, 2, 1);  
  System.out.println(isPalindrome(list));  
}
```

```
}
```

Dry Run

left	right	match?
0	4	1==1 yes
1	3	2==2 yes
2	2	left==right, stop

Output: true

Question 15: Group ArrayList Elements by Even/Odd

Problem

Group [1, 2, 3, 4, 5, 6] into two lists: even numbers and odd numbers.

Pseudocode

```
FUNCTION groupByParity(list):  
  evens = empty ArrayList, odds = empty ArrayList  
  FOR each v in list:  
    IF v MOD 2 == 0: ADD v to evens  
    ELSE: ADD v to odds  
  RETURN (evens, odds)
```

Java Code

```
static void groupByParity(List<Integer> list) {  
  List<Integer> evens = new ArrayList<>();  
  List<Integer> odds = new ArrayList<>();  
  for (int v : list) {  
    if (v % 2 == 0) evens.add(v); else odds.add(v);  
  }  
  System.out.println("Evens=" + evens + " Odds=" + odds);  
}  
public static void main(String[] args) {  
  List<Integer> list = List.of(1, 2, 3, 4, 5, 6);  
  groupByParity(list);  
}
```

Dry Run

v	v%2==0?	goes to
1	no	odds
2	yes	evens
3	no	odds
4	yes	evens
5	no	odds
6	yes	evens

Output: Evens=[2, 4, 6] Odds=[1, 3, 5]

Question 16: Find the Frequency of Each Element in an ArrayList

Problem

Count how many times each element appears in [1, 2, 2, 3, 3, 3, 4].

Pseudocode

```
FUNCTION frequency(list):  
  map = empty HashMap  
  FOR each v in list:  
    map[v] = map.getOrDefault(v, 0) + 1  
  RETURN map
```

Java Code

```
static Map<Integer, Integer> frequency(List<Integer> list) {  
  Map<Integer, Integer> map = new LinkedHashMap<>();  
  for (int v : list) {  
    map.put(v, map.getOrDefault(v, 0) + 1);  
  }  
  return map;  
}  
  
public static void main(String[] args) {  
  List<Integer> list = List.of(1, 2, 2, 3, 3, 3, 4);  
  System.out.println(frequency(list));  
}
```

Dry Run

v	map after
1	{1:1}
2	{1:1, 2:1}
2	{1:1, 2:2}
3	{1:1, 2:2, 3:1}
3	{1:1, 2:2, 3:2}
3	{1:1, 2:2, 3:3}
4	{1:1, 2:2, 3:3, 4:1}

Output: {1=1, 2=2, 3=3, 4=1}

Question 17: Remove an Element from an ArrayList While Iterating (Iterator)

Problem

From [10, 15, 20, 25, 30], remove all multiples of 10 safely while iterating.

Pseudocode

```
FUNCTION removeMultiplesOf10(list):  
  it = list.iterator()  
  WHILE it.hasNext():  
    v = it.next()  
    IF v MOD 10 == 0: it.remove()
```

Java Code

```
static void removeMultiplesOf10(List<Integer> list) {  
  Iterator<Integer> it = list.iterator();  
  while (it.hasNext()) {  
    int v = it.next();  
    if (v % 10 == 0) it.remove();  
  }  
}  
  
public static void main(String[] args) {  
  List<Integer> list = new ArrayList<>(List.of(10, 15, 20, 25, 30));  
  removeMultiplesOf10(list);  
  System.out.println(list);  
}
```

Dry Run

v	v%10==0?	action
10	yes	it.remove()
15	no	keep
20	yes	it.remove()
25	no	keep
30	yes	it.remove()

Output: [15, 25]

Question 18: Find the Union of Two ArrayLists

Problem

Find the union (all distinct elements) of [1, 2, 3] and [3, 4, 5].

Pseudocode

```
FUNCTION union(list1, list2):  
  result = new LinkedHashSet from list1  
  ADD all of list2 to result  
  RETURN new ArrayList from result
```

Java Code

```
static List<Integer> union(List<Integer> list1, List<Integer> list2) {  
  Set<Integer> result = new LinkedHashSet<>(list1);  
  result.addAll(list2);  
  return new ArrayList<>(result);  
}  
  
public static void main(String[] args) {  
  List<Integer> a = List.of(1, 2, 3), b = List.of(3, 4, 5);  
  System.out.println(union(a, b));  
}
```

Dry Run

Step	result after
from list1	{1,2,3}
add list2 (3 already present)	{1,2,3,4,5}

Output: [1, 2, 3, 4, 5]

Question 19: Find the Kth Smallest Element in an ArrayList

Problem

Find the 3rd smallest element in [9, 2, 7, 4, 1, 8].

Pseudocode

```
FUNCTION kthSmallest(list, k):  
  sorted = copy of list, sorted ascending  
  RETURN sorted.get(k - 1)
```

Java Code

```
static int kthSmallest(List<Integer> list, int k) {  
  List<Integer> sorted = new ArrayList<>(list);  
  Collections.sort(sorted);  
  return sorted.get(k - 1);  
}  
public static void main(String[] args) {  
  List<Integer> list = List.of(9, 2, 7, 4, 1, 8);  
  System.out.println(kthSmallest(list, 3));  
}
```

Dry Run

Step	Value
sorted list	[1, 2, 4, 7, 8, 9]
k=3 -> index k-1=2	sorted.get(2) = 4

Output: 4

Question 20: Flatten a 2D ArrayList into a 1D ArrayList

Problem

Flatten [[1, 2], [3, 4], [5]] into a single ArrayList [1, 2, 3, 4, 5].

Pseudocode

```
FUNCTION flatten(grid):  
  result = empty ArrayList  
  FOR each row in grid:  
    FOR each v in row:  
      ADD v to result  
  RETURN result
```

Java Code

```
static List<Integer> flatten(List<List<Integer>> grid) {  
  List<Integer> result = new ArrayList<>();  
  for (List<Integer> row : grid) {  
    result.addAll(row);  
  }  
  return result;  
}  
public static void main(String[] args) {  
  List<List<Integer>> grid = List.of(List.of(1, 2), List.of(3, 4), List.of(5));  
  System.out.println(flatten(grid));  
}
```

```
}
```

Dry Run

row	result after addAll
[1, 2]	[1, 2]
[3, 4]	[1, 2, 3, 4]
[5]	[1, 2, 3, 4, 5]

Output: [1, 2, 3, 4, 5]

End of Practice Set — 20 Questions Covering ArrayList.